LAB ASSIGNMENT – 5 (Indexing & Sorting)
DATE – 16.02.2026

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ROLL NO. : UG/02/BTCSECSF/2023/013

SUBJECT : Unstructured Database Lab
SEC: E

Schema

{ emp_id, name, department, designation, city, salary, experience}

Initial Data

Emp ID	Name	Department	Designation	City	Salary	Experience
101	Amit	IT	Developer	Kolkata	60000	3
102	Neha	HR	HR Executive	Delhi	45000	2
103	Ravi	IT	Tester	Mumbai	50000	4
104	Pooja	Finance	Accountant	Chennai	55000	5
105	Karan	IT	Developer	Kolkata	65000	6
106	Sneha	HR	Manager	Delhi	70000	7
107	Arjun	Finance	Analyst	Mumbai	48000	2
108	Meera	IT	Developer	Bangalore	75000	8
109	Rohit	IT	Team Lead	Pune	90000	10
110	Anjali	HR	Recruiter	Delhi	42000	1

Lab Questions:

Q1. Display all employee records.

```
test> use companyDB
switched to db companyDB
```

```
companyDB> db.employees.insertMany([
... {emp_id:101,name:"Amit",department:"IT",designation:"Developer",city:"Kolkata",salary:60000,experience:3},
... {emp_id:102,name:"Neha",department:"HR",designation:"HR Executive",city:"Delhi",salary:45000,experience:2},
... {emp_id:103,name:"Ravi",department:"IT",designation:"Tester",city:"Mumbai",salary:50000,experience:4},
... {emp_id:104,name:"Pooja",department:"Finance",designation:"Accountant",city:"Chennai",salary:55000,experience:5},
... {emp_id:105,name:"Karan",department:"IT",designation:"Developer",city:"Kolkata",salary:65000,experience:6},
... {emp_id:106,name:"Sneha",department:"HR",designation:"Manager",city:"Delhi",salary:70000,experience:7},
... {emp_id:107,name:"Arjun",department:"Finance",designation:"Analyst",city:"Mumbai",salary:48000,experience:2},
... {emp_id:108,name:"Meera",department:"IT",designation:"Developer",city:"Bangalore",salary:75000,experience:8},
... {emp_id:109,name:"Rohit",department:"IT",designation:"Team Lead",city:"Pune",salary:90000,experience:10},
... {emp_id:110,name:"Anjali",department:"HR",designation:"Recruiter",city:"Kolkata",salary:42000,experience:1}
... ])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('6992bf2ffaece55da81e2621'),
    '1': ObjectId('6992bf2ffaece55da81e2622'),
    '2': ObjectId('6992bf2ffaece55da81e2623'),
    '3': ObjectId('6992bf2ffaece55da81e2624'),
    '4': ObjectId('6992bf2ffaece55da81e2625'),
    '5': ObjectId('6992bf2ffaece55da81e2626'),
    '6': ObjectId('6992bf2ffaece55da81e2627'),
    '7': ObjectId('6992bf2ffaece55da81e2628'),
    '8': ObjectId('6992bf2ffaece55da81e2629'),
    '9': ObjectId('6992bf2ffaece55da81e262a')
  }
}
```

```
companyDB> db.employees.find()
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
```

```
    _id: ObjectId('6992bf2ffaece55da81e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
```

```
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q2. Create an index on emp_id field.

```
companyDB> db.employees.createIndex({emp_id:1})
emp_id_1
```

Q3. Create an index on department field.

```
companyDB> db.employees.createIndex({department:1})
department_1
```

Q4. Create a compound index on department and salary.

```
companyDB> db.employees.createIndex({department:1, salary:-1})
department_1_salary_-1
```

Q5. Find employees working in IT department.

```
companyDB> db.employees.find({department:"IT"})
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
```

```
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
```

Q6. Display employees whose salary is greater than 60000.

```
companyDB> db.employees.find({salary:{$gt:60000}})
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
```

```
{
  _id: ObjectId('6992bf2ffaece55da81e2629'),
  emp_id: 109,
  name: 'Rohit',
  department: 'IT',
  designation: 'Team Lead',
  city: 'Pune',
  salary: 90000,
  experience: 10
}
```

Q7.Sort employees by salary in descending order.

```
companyDB> db.employees.find().sort({salary:-1})
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
```

```
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2621'),
    emp_id: 101,
    name: 'Amit',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 60000,
    experience: 3
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2624'),
    emp_id: 104,
    name: 'Pooja',
    department: 'Finance',
    designation: 'Accountant',
    city: 'Chennai',
    salary: 55000,
    experience: 5
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2623'),
    emp_id: 103,
    name: 'Ravi',
    department: 'IT',
    designation: 'Tester',
    city: 'Mumbai',
    salary: 50000,
    experience: 4
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2627'),
    emp_id: 107,
    name: 'Arjun',
    department: 'Finance',
    designation: 'Analyst',
```

```
    designation: 'Analyst',
    city: 'Mumbai',
    salary: 48000,
    experience: 2
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2622'),
    emp_id: 102,
    name: 'Neha',
    department: 'HR',
    designation: 'HR Executive',
    city: 'Delhi',
    salary: 45000,
    experience: 2
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e262a'),
    emp_id: 110,
    name: 'Anjali',
    department: 'HR',
    designation: 'Recruiter',
    city: 'Kolkata',
    salary: 42000,
    experience: 1
  }
]
```

Q8. Display top 3 highest paid employees.

```
companyDB> db.employees.find().sort({salary:-1}).limit(3)
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  }
]
```

Q9. Display employees having experience more than 5 years.

.

```
companyDB> db.employees.find({experience:{$gt:5}})
[
  {
    _id: ObjectId('6992bf2ffaece55da81e2625'),
    emp_id: 105,
    name: 'Karan',
    department: 'IT',
    designation: 'Developer',
    city: 'Kolkata',
    salary: 65000,
    experience: 6
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2626'),
    emp_id: 106,
    name: 'Sneha',
    department: 'HR',
    designation: 'Manager',
    city: 'Delhi',
    salary: 70000,
    experience: 7
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2628'),
    emp_id: 108,
    name: 'Meera',
    department: 'IT',
    designation: 'Developer',
    city: 'Bangalore',
    salary: 75000,
    experience: 8
  },
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
```

```
  {
    _id: ObjectId('6992bf2ffaece55da81e2629'),
    emp_id: 109,
    name: 'Rohit',
    department: 'IT',
    designation: 'Team Lead',
    city: 'Pune',
    salary: 90000,
    experience: 10
  }
]
```

Q10. Analyze query performance using explain().

```
companyDB> db.employees.find({department:"IT"}).explain("executionStats")
{
  explainVersion: '1',
  queryPlanner: {
    namespace: 'companyDB.employees',
    parsedQuery: { department: { '$eq': 'IT' } },
    indexFilterSet: false,
    queryHash: 'EB9C5E66',
    planCacheShapeHash: 'EB9C5E66',
    planCacheKey: '25CB5D62',
    optimizationTimeMillis: 0,
    maxIndexedOrSolutionsReached: false,
    maxIndexedAndSolutionsReached: false,
    maxScansToExplodeReached: false,
    prunedSimilarIndexes: false,
    winningPlan: {
      isCached: false,
      stage: 'FETCH',
      inputStage: {
        stage: 'IXSCAN',
        keyPattern: { department: 1 },
        indexName: 'department_1',
        isMultiKey: false,
        multiKeyPaths: { department: [] },
        isUnique: false,
        isSparse: false,
        isPartial: false,
        indexVersion: 2,
        direction: 'forward',
        indexBounds: { department: [ '["IT", "IT"]' ] }
      }
    },
    rejectedPlans: [
      {
        isCached: false,
        stage: 'FETCH',
        inputStage: {
          stage: 'IXSCAN',
```